

SECTION B

Option A – Alternating Currents

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	Reactance (accept impedance/resistance) of inductor increases with frequency (1) Resistor takes a smaller portion of pd (OR current decreases so pd across R decreases) (1)	2			2		
		(ii)	Equating $\omega L = R$ (1) Answer = 141 k[Hz] (1)		2		2	2	
		(iii)	Correct method (implied by correct answer) (1) Answer = 1.77 [V] (1.25 [V] no marks) (1)		2		2	1	
	(b)	(i)	Flux proportional to B (1) Flux proportional to A (1) Same flux for each turn (N) (1) OR using Faraday's e.g. $V = \frac{d}{dt}(BAN \cos \theta)$ (1) Taking B and A outside bracket (1) Stating B and A are constants (1) OR Faraday's law stated (1) Correct explanation of rate of change of flux with reference to cutting of field lines, linked to B (1) Correct explanation of rate of change of flux with reference to cutting of field lines, linked to A (1)	3			3		
		(ii)	When flat (or wtte) because cutting at right angles OR flat because max rate of change of flux	1			1		